A220 Profiles

Revision 8.2

Breeze

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BTA

Normal Takeoff

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items



FMA setup with proper modes selected (NAV Departure)

NOTE: NAV will not be selected for Non-RNAV coded departures

When thrust mode is active

Thrust

Check

Autothrottle not available or disconnected

Set Thrust

Thrust Set

80 Knots Thrust Set

Check

Autocallout

V1

Rotate

When a positive rate of climb is confirmed

Positive Rate

Gear Up

Gear Up

Confirm/select HDG or NAV

400 ft

HDG or FMS 1(2)

Check

Confirm OR select flight level change

(V)FLC

Check

Retract Flaps on Schedule

Flap 3/2/1

Speed Checked, Flap #

Flap 0, After Takeoff Checklist

Speed Checked, Flap 0 After Takeoff Checklist Complete

Acceleration Altitude

Stabilized Climb

Autopilot ON

If desired



Takeoff Begins

80 knots & at required N1 value

V1

VR

Normal Takeoff FMA Indications

Autothrottle armed

THRUST	TO FMS1	← AT	VTO VALTS	VFLC	

Autothrottle engaged

THRUST SPD	TO FMS1	← AT	VTO VALTS	VFLC	

80 kts

HOLD SPD	TO FMS1	← AT	VTO VALTS	VFLC	

The Auto Throttle system must be armed prior to the commencement of a takeoff roll if conducting a derated (TO-1, TO-2, TO-3) or reduced thrust (flex) takeoff. A rejected takeoff is required if the auto throttle system is not armed prior to the commencement of a derated or reduced thrust takeoff.

For full thrust (TO) takeoffs the auto throttle system is not required. Crews may refer to Chapter 6 Reduced Thrust Takeoff limitation for further information.

NOTE: Non–RNAV departures

400' ("Heading")

THRUST SPD	HDG	← AP AT	VTO VALTS	VFLC	

1000 or accel altitude ("(V)FLCH")

THRUST	HDG	← AP AT	VFLC VALTS		

NOTE: RNAV coded departure

400' ("FMS 1 (2)")

THRUST SPD	FMS1	← AP AT	VTO VALTS	VFLC	

1000 or accel altitude ("(V)FLC)")

THRUST	FMS1	← AP AT	VFLC VALTS		

NOTE: Takeoff Path: NO turns shall be commenced below 1,000 ft AFE when takeoff weather is less than 1,000 ft ceiling and 3sm visibility unless a Special Departure Procedure prescribes otherwise or the assigned instrument departure procedure specifically requires a turn before reaching 1,000 ft AGL

High Wind Takeoff

Pilot Flying Callouts

Pilot Monitoring Callouts

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For high wind assessment see FCOM2 Chapter 3

Set thrust at 55% N1 and allow to stabilize.

Upon reaching 20 knots ground speed set takeoff thrust.
Failure to advance thrust levers to the takeoff setting by 60 knots ground speed may cause the extension of spoilers.

See AeroData uplink report for "High Wind Takeoff Procedure Required" note



FMA setup with proper modes selected (NAV Departure)

NOTE: NAV will not be selected for Non-RNAV coded departures

For HIGH WINDS only

20 knots Ground Speed

Autothrottle not available or disconnected

Set Thrust

Thrust Set

When thrust mode is active

Thrust

Check

80 Knots Thrust Set

Check

Autocallout

V1

Rotate

When a positive rate of climb is confirmed

Positive Rate

Gear Up

Gear Up

Confirm/select HDG or NAV

400 ft

HDG or FMS 1(2)

Check

Confirm OR select flight level change

(V)FLC

Check

Retract flaps on schedule

Flap 3/2/1

Speed Checked, Flap #

Flap 0, After Takeoff Checklist

Speed Checked, Flap 0 After Takeoff Checklist Complete

Acceleration Altitude

Stabilized Climb

Autopilot ON

If desired

Takeoff Begins

80 knots & at required N1 value

V1

VR

Normal Takeoff FMA Indications

Autothrottle armed

THRUST	TO FMS1	AT	VTO VALTS VFLC

Autothrottle engaged

THRUST SPD	TO FMS1	AT	VTO VALTS VFLC

80 kts

HOLD SPD	TO FMS1	AT	VTO VALTS VFLC

The Auto Throttle system must be armed prior to the commencement of a takeoff roll if conducting a derated (TO-1, TO-2, TO-3) or reduced thrust (flex) takeoff. A rejected takeoff is required if the auto throttle system is not armed prior to the commencement of a derated or reduced thrust takeoff.

For full thrust (TO) takeoffs the auto throttle system is not required. Crews may refer to Chapter 6 Reduced Thrust Takeoff limitation for further information.

NOTE: Non–RNAV departures

400' ("Heading")

THRUST SPD	HDG	AP AT	VTO VALTS VFLC

1000 or accel altitude ("(V)FLCH")

THRUST	HDG	AP AT	VFLC VALTS

NOTE: RNAV coded departure

400' ("FMS 1 (2)")

THRUST SPD	FMS1	AP AT	VTO VALTS VFLC

1000 or accel altitude ("(V)FLC")

THRUST	FMS1	AP AT	VFLC VALTS

NOTE: Takeoff Path: NO turns shall be commenced below 1,000 ft AFE when takeoff weather is less than 1,000 ft ceiling and 3sm visibility unless a Special Departure Procedure prescribes otherwise or the assigned instrument departure procedure specifically requires a turn before reaching 1,000 ft AGL

Engine Failure After V1

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Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items

When a positive rate of climb is confirmed

Positive Rate

Gear Up

Gear Up

When thrust mode is active

Thrust

Check

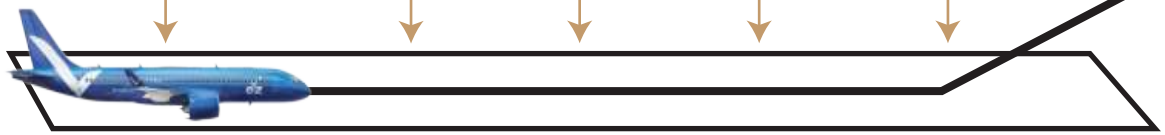
80 Knots Thrust Set

Check

V1

Engine Failure

Rotate



Takeoff Begins

80 knots

V1

Loss of thrust

VR

Autothrottle armed

THRUST	TO FMS1	AT	VTO VALTS	VFLC

Autothrottle engaged

THRUST SPD	TO FMS1	AT	VTO VALTS	VFLC

80 kts

HOLD SPD	FMS1	AT	VTO VALTS	VFLC

If turns are required in HDG mode, HDG 1/2 bank must be used through cleanup until reaching green dot.

CONTINUED...

Acceleration Altitude Note: Green dot speed may be above MAX Man speed, keep selected speed above green dot once visible.

Acceleration Altitude

Confirm/select HDG or NAV

Select MAN speed, set speed MAX Man, then select ALT

400 ft

HDG or FMS 1(2)

Check

(V)ALT

Check

Retract flaps on schedule

Flap #

Speed Checked, Flap #

Approaching green dot: When speed trend reaches green dot, select FLC and adjust speed to green dot

(V) FLC

Check

After Takeoff Checklist

After Takeoff Checklist Complete

OR

Call for appropriate checklist

Checklist Priority

Red

White

Yellow

400' ("FMS 1 (2)")

THRUST SPD	FMS1	AP AT	VTO VALTS	VFLC

1000 or accel altitude ("(V)ALT")

SPD	FMS1	AP AT	VALT

400' ("Heading")

THRUST SPD	HDG	AP AT	VTO VALTS	VFLC

1000 or accel altitude ("(V)ALT")

SPD	HDG	AP AT	VALT

400' (stabilized climb)

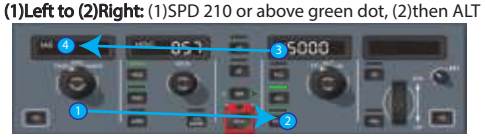
Autopilot ON

If desired

1000'

(Or acceleration altitude)

(1)Left to (2)Right: (1)SPD 210 or above green dot, (2)then ALT



(3) Right to (4) Left: (3)FLC then (4)set speed green dot



Final climb Segment

It is important to maintain the speed target above the current speed of the aircraft to avoid unwanted thrust reduction (AT ON) in the level acceleration segment.

Rejected Takeoff

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Pilot Flying Callouts

Pilot Monitoring Callouts

Captain Callouts

Dashed boxes represent Action Items

Captains will accomplish the rejected takeoff procedure unless Captain incapacitation, then the First Officer will perform all duties and tasks.

When thrust mode is active

Thrust

Check

80 Knots Thrust Set

Check

At the first sign of emergency abnormality prior to V1

Reject

Reverse-green or no reverse 1 (2)

Advise tower

When stopped on the runway:

1. Throttles idle
2. Set parking brake
3. PA

This is the Captain, remain seated, remain seated



Takeoff Begins

80 knots

Captain takes controls: unless incapacitated

Decelerating

Stopped on runway with parking brake set

Assess situation and accomplish **applicable checklist** items:
Order emergency evacuation **ONLY IF NECESSARY**

GOM 5.21

If an engine failure occurs during takeoff prior to V1 or if the takeoff is aborted for any reason regardless of which pilot is flying, the Captain will immediately assume full control of the aircraft.

ILS Approach: Vectored Above FAF Altitude

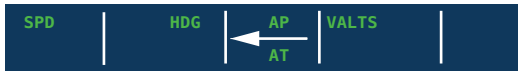
Breeze

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items

Breeze A220



Crew being given vectors for ILS approach



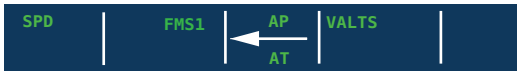
To enhance the LOC capture (minimum overshoot, in the LOC course direction), the flight crew should use the FMS guidance NAV mode and delay the selection of the APPR mode.

Cleared for the approach:

1. Arm FMS1(2) with NAV



Crew is cleared for the ILS approach on a heading to intercept

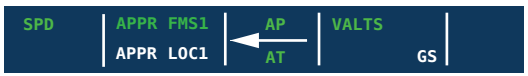


When LOC deviation is 1 dot or less

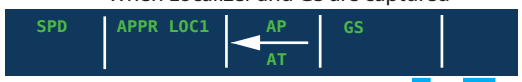
NOTE: The flight crew should select APPR when all of the following conditions are applicable:

1. Cleared for the approach
2. Distance to RWY 25 miles or less
3. LOC deviation is 1 dot or less





When Localizer and GS are captured



Minimum 3nm to FAF: Set SPD MAN and reduce speed to VAPP

Flaps 4(5), Before Landing Checklist

Speed Checked, Flaps 4 (5) Before Landing Checklist

Accomplish Before Landing Checklist

"Approach Lights" = "Continuing"
"Runway" = "Landing"

Note: Pilot monitoring will announce "1,000" and "500" **if no autocallout**

When FMS inbound course is captured

Approach FMS 1(2)

When localizer is captured

Approach LOC 1(2)

When glideslope captured

Glideslope



Blue dashed line represents NAV to NAV Transfer as seen on HSI

First positive inward motion of final approach course

Course Alive

Flaps on F-speeds

Flaps 1

Speed Checked Flaps 1

Flaps on F-speeds

Flaps 2

Speed Checked Flaps2

Minimum of 10nm from FAF

Minimum 5nm to FAF

Flaps on F-speeds

Gear down Flaps 3

Gear Down Speed Checked Flaps 3

Missed approach altitude can be set after glideslope capture

Missed Approach Altitude Set ____

____ Set

Below 2,000 ft AGL

LOC 1 (2)

FAF [name] Altitude

Checked

1,000ft AFE

Stable or Unstable Go Around

Stable or Unstable Go Around

500 ft AFE

Stable or Unstable Go Around

Intermediate Fix

Note: Aircraft may prioritize next published altitude over constant angle descent when inside the penultimate fix.

Note: Configuration distances are a recommendation only to aid in training

ILS Approach: At FAF Altitude

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items

Breeze A220



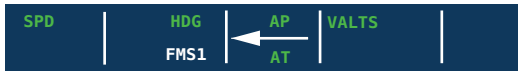
Crew being given vectors for ILS approach

Cleared for the approach:

1. Arm FMS1(2) with NAV



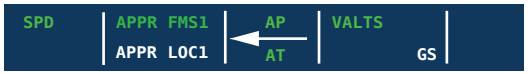
To enhance the LOC capture (minimum overshoot, in the LOC course direction), the flight crew should use the FMS guidance NAV mode and delay the selection of the APPR mode.



Crew is cleared for the ILS approach on a heading to intercept



When LOC deviation is 1 dot or less



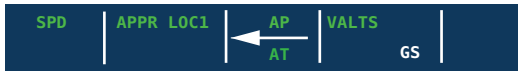
NOTE: The flight crew should select APPR when all of the following conditions are applicable:

1. Cleared for the approach
2. Distance to RWY 25 miles or less
3. LOC deviation is 1 dot or less



"Approach Lights" = "Continuing"
"Runway" = "Landing"

Note: Pilot monitoring will announce "1,000" and "500" **if no autocallout**



When localizer is captured

Approach LOC 1(2)

1 dot below glideslope

Set SPD MAN and reduce speed to VAPP

Flaps 4(5), Before Landing Checklist

Speed Checked, Flaps 4 (5) Before Landing Checklist

Accomplish Before Landing Checklist

Missed approach altitude can be set after glideslope capture OR When VALTS prior to FAF

Missed Approach Altitude Set ____

____ Set



Blue dashed line represents NAV to NAV Transfer as seen on HSI

First positive inward motion of final approach course

Course Alive

Flaps on F-speeds

Flaps 1

Speed Checked Flaps 1

Flaps on F-speeds

Flaps 2

Speed Checked Flaps2

Minimum of 10nm from FAF

Flaps on F-speeds

Gear down Flaps 3

Gear Down Speed Checked Flaps 3

2 dots below glideslope

FAF [name] Altitude

Checked

LOC 1 (2)

1,000ft AFE

Stable or Unstable Go Around

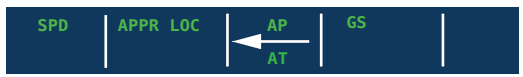
Stable or Unstable Go Around

500 ft AFE

Stable or Unstable Go Around

When glideslope captured

Glideslope





Note: Configuration distances are a recommendation only to aid in training

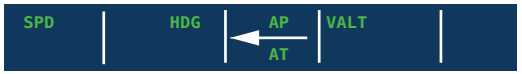
RNAV Approach: Vectored Above FAF Altitude

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items

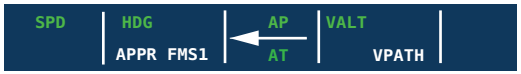


Crew being given vectors for RNAV approach

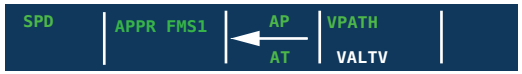
Cleared for the approach:

1. Arm approach (APPR)
2. Arm VNAV (if not already)
3. Set FAF altitude, **once established**, e.g. 1800'

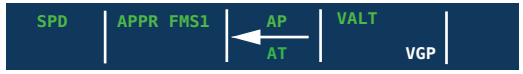




Crew is cleared for the RNAV approach on a heading to intercept and FAF altitude is set



VPATH engages and descends to next published altitude (see **NOTE**) and FMS course captured



Aircraft levels off at published altitude



VGP is captured

"Approach Lights" = "Continuing"
"Runway" = "Landing"

Note: Pilot monitoring will announce "1,000" and "500" **if no autocallout**

When the FMS inbound course is captured

Approach FMS 1(2)



First positive inward motion of final approach course

Course Alive

Flaps on F-speeds

Flaps 1

Speed Checked Flaps 1

Flaps on F-speeds

Flaps 2

Speed Checked Flaps2

Minimum of 10nm from FAF

When VGP is captured

VGP

Missed approach altitude can be set after glidepath capture

Missed Approach Altitude Set ____

____ Set

Minimum 5nm to FAF
Flaps on F-speeds

Gear down Flaps 3

Gear Down Speed Checked Flaps 3

Minimum 3nm to FAF: Set SPD MAN and reduce speed to VAPP

Flaps 4(5), Before Landing Checklist

Speed Checked, Flaps 4 (5) Before Landing Checklist

Accomplish Before Landing Checklist

FAF [name] Altitude

Checked

1,000ft AFE

Stable or Unstable Go Around

Stable or Unstable Go Around

500 ft AFE

Stable or Unstable Go Around

Note: Aircraft may prioritize next published altitude over constant angle descent when inside the penultimate fix.

Intermediate Fix

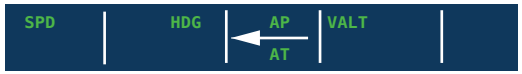
RNAV Approach: At FAF Altitude

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items

Breeze A220

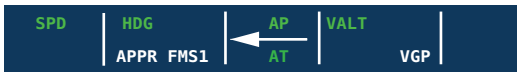


Crew being given vectors for RNAV approach

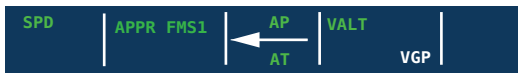
Cleared for the approach at the FAF altitude:

1. Arm approach (APPR)
2. Arm VNAV (if not already)





Crew is cleared for the RNAV approach on a heading to intercept and FAF altitude is set



When the FMS inbound course is captured

1 dot below glidepath

Set SPD MAN and reduce speed to VAPP

Flaps 4(5), Before Landing Checklist

Speed Checked, Flaps 4 (5) Before Landing Checklist

Accomplish Before Landing Checklist

"Approach Lights" = "Continuing"
"Runway" = "Landing"

Note: Pilot monitoring will announce "1,000" and "500" **if no autocallout**

When FMS inbound course is captured

Approach FMS 1(2)

Missed approach altitude can be set after glidepath capture

Missed Approach Altitude Set ____

____ Set



First positive inward motion of final approach course

Course Alive

Flaps on F-speeds

Flaps 1

Speed Checked Flaps 1

Flaps on F-speeds

Flaps 2

Speed Checked Flaps2

Minimum of 10nm from FAF

Flaps on F-speeds

Gear down Flaps 3

Gear Down Speed Checked Flaps 3

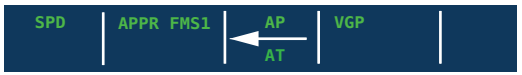
2 dots below glidepath

FAF [name] Altitude

Checked

When VGP is captured

VGP



1,000ft AFE

Stable or Unstable Go Around

Stable or Unstable Go Around

500 ft AFE

Stable or Unstable Go Around



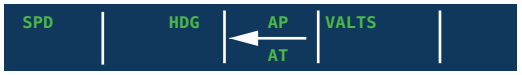
LOC Approach: Vectored Above FAF Altitude

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items



Crew being given vectors for LOC approach

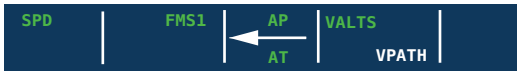
Cleared for the approach:

1. Arm FMS1(2) with NAV
2. Arm VNAV (if not already)
3. Set DDA, **once established**, e.g. 810'



MDA 810 Set | 810 Set





Crew is cleared for the LOC approach on a heading to intercept and DDA is set

To enhance the LOC capture (minimum overshoot, in the LOC course direction), the flight crew should use the FMS guidance NAV mode and delay the selection of the APPR mode.

NOTE: The flight crew should select APPR when all of the following conditions are applicable:

1. Cleared for the approach
2. Distance to RWY 25 miles or less
3. LOC deviation is 1 dot or less



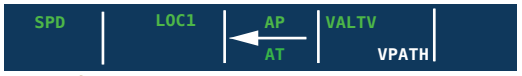
DDA: Pilots must calculate and use Derived Decision Altitude (DDA) by adding 50 ft to the MDA. This is to ensure that the aircraft does not descend below the MDA during the go-around maneuver.

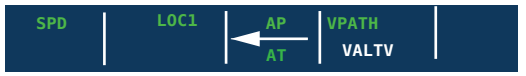




When LOC deviation is 1 dot or less arm the APPR for NAV to NAV transfer







VNAV engages and descends to next published altitude



When FMS inbound course is captured

Approach FMS 1(2)



Blue dashed line represents NAV to NAV Transfer as seen on HSI

When localizer is captured: Select NAV



LOC 1(2)

Minimum 3nm to FAF: Set SPD MAN and reduce speed to VAPP

Flaps 4(5), Before Landing Checklist

Speed Checked, Flaps 4 (5) Before Landing Checklist

Accomplish Before Landing Checklist

"Approach Lights" = "Continuing"
"Runway" = "Landing"

Note: Pilot monitoring will announce "1,000" and "500" **if no autocallout**

First positive inward motion of final approach course

Course Alive

Flaps on F-speeds

Flaps 1

Speed Checked Flaps 1

Magenta dashed line represents vertical navigation (VNAV) until deselected

Flaps on F-speeds

Flaps 2

Speed Checked Flaps2

Minimum of 10nm from FAF

Minimum 5nm to FAF
Flaps on F-speeds

Gear down Flaps 3

Gear Down Speed Checked Flaps 3

Once established on published descent angle but before DDA

Missed Approach Altitude Set ____

____ Set

On path:
Deselect VNAV

FPA

Set charted angle

FAF [name] Altitude

Checked

1,000ft AFE

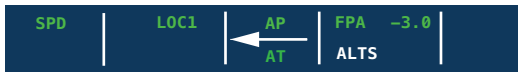
Stable or Unstable Go Around

Stable

500 ft AFE

Stable or Unstable Go Around

Intermediate Fix





Note: Aircraft may prioritize next published altitude over constant angle descent when inside the penultimate fix.

Note: Configuration distances are a recommendation only to aid in training

LOC Approach: At FAF Altitude

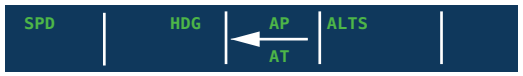
Breeze

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items



Crew being given vectors for LOC approach

To enhance the LOC capture (minimum overshoot, in the LOC course direction), the flight crew should use the FMS guidance NAV mode and delay the selection of the APPR mode.

Cleared for the approach:

1. Arm FMS1(2) with NAV
2. Set DDA, **once established**, e.g. 810'



MDA 810 Set

810 Set





Crew is cleared for the LOC approach on a heading to intercept and DDA is set

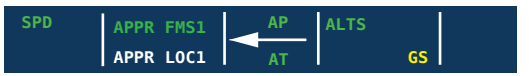
NOTE: The flight crew should select APPR when all of the following conditions are applicable:

1. Cleared for the approach
2. Distance to RWY 25 miles or less
3. LOC deviation is 1 dot or less



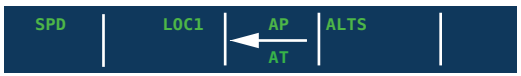
"Approach Lights" = "Continuing"
"Runway" = "Landing"

Note: Pilot monitoring will announce "1,000" and "500" **if no autocallout**



When LOC deviation is 1 dot or less arm the APPR for NAV to NAV transfer





When localizer is captured:

Select NAV



LOC 1(2)

Minimum 3nm to FAF: Set SPD MAN and reduce speed to VAPP

Flaps 4(5), Before Landing Checklist

Speed Checked, Flaps 4 (5) Before Landing Checklist

Accomplish Before Landing Checklist

Deselect VNAV if engaged

Missed approach altitude can be set once established on charted descent angle

Missed Approach Altitude Set ____

____ Set



Blue dashed line represents NAV to NAV Transfer as seen on HSI

First positive inward motion of final approach course

Course Alive

Flaps on F-speeds

Flaps 1

Speed Checked Flaps 1

Flaps on F-speeds

Flaps 2

Speed Checked Flaps2

Minimum of 10nm from FAF

Minimum 5nm to FAF

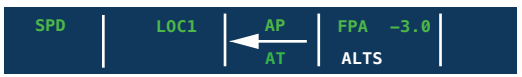
Flaps on F-speeds

Gear down Flaps 3

Gear Down Speed Checked Flaps 3

. 3 nm prior to FAF: FPA set to charted angle

FPA



FAF [name] Altitude

Checked

LOC 1 (2)

1,000ft AFE

Stable or Unstable Go Around

Stable or Unstable Go Around

500 ft AFE

Stable or Unstable Go Around



Note: Configuration distances are a recommendation only to aid in training

VOR Approach: Vectored Above the FAF

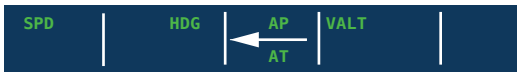
Breeze

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items

Breeze A220



Crew being given vectors for VOR approach

FCOM 2: Chapter 3

VOR (GPS/GNSS) overlay approaches are flown using the FMS as the primary source of navigation

For VOR approaches without GPS/GNSS overlay, manually tune the correct VOR frequency on NAV 1 and NAV 2

If Arrival Data tab populates "**APPR FOR REF ONLY**," then BRG pointers must be used as a means of displaying raw VOR data.

Cleared for the approach:

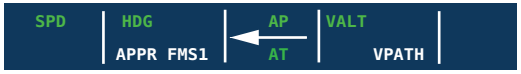
1. Arm approach (APPR)
2. Arm VNAV (if not already)
3. Set DDA altitude, **once established**, e.g. 810'



MDA 810 Set

810 Set

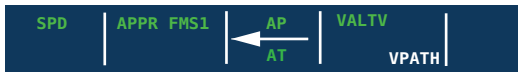




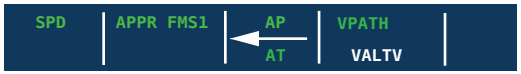
Crew is cleared for the VOR approach on a heading to intercept and DDA altitude is set



VPATH engages and descends to next published altitude (see **NOTE**) and FMS course captured



Aircraft levels off at published altitude



VPATH engages

When the FMS inbound course is captured

Approach FMS 1(2)



First positive inward motion of final approach course

Course Alive

Flaps on F-speeds

Flaps 1

Speed Checked Flaps 1

Flaps on F-speeds

Flaps 2

Speed Checked Flaps2

Minimum of 10nm from FAF

Minimum 3nm to FAF: Set SPD MAN and reduce speed to VAPP

Flaps 4(5), Before Landing Checklist

Speed Checked, Flaps 4 (5) Before Landing Checklist

Accomplish Before Landing Checklist

Minimum 5nm to FAF Flaps on F-speeds

Gear down Flaps 3

Gear Down Speed Checked Flaps 3

On path: **Deselect VNAV**

FPA

Set charted angle

Before FAF (below MAA)

Missed Approach Altitude Set ____

____ Set

FAF [name] Altitude

Checked

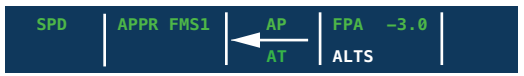
1,000ft AFE

Stable or Unstable Go Around

Stable

500 ft AFE

Stable or Unstable Go Around



Intermediate Fix



"Approach Lights" = "Continuing"
"Runway" = "Landing"

Note: Pilot monitoring will announce "1,000" and "500" **if no autocallout**

Note: Aircraft may prioritize next published altitude over constant angle descent when inside the penultimate fix.

DDA: Pilots must calculate and use Derived Decision Altitude (DDA) by adding 50 ft to the MDA. This is to ensure that the aircraft does not descend below the MDA during the go-around maneuver.

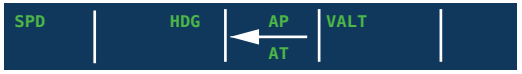
VOR Approach: At FAF Altitude

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items



Crew being given vectors for RNAV approach

Cleared for the approach at the FAF Altitude:

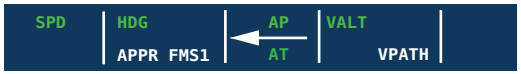
1. Arm approach (APPR)
2. Set DDA, e.g. 810



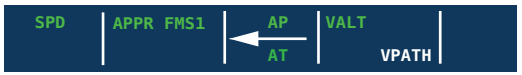
MDA 810 Set

810 Set





Cleared for the VOR approach on a heading to intercept and DDA altitude is set (with VNAV armed).



FMS inbound course is captured with DDA set in altitude selector and VNAV armed.

FCOM 2: Chapter 3

VOR (GPS/GNSS) overlay approaches are flown using the FMS as the primary source of navigation

For VOR approaches without GPS/GNSS overlay, manually tune the correct VOR frequency on NAV 1 and NAV 2

If Arrival Data tab populates "**APPR FOR REF ONLY**," then BRG pointers must be used as a means of displaying raw VOR data.

"Approach Lights" = "Continuing"
"Runway" = "Landing"

Note: Pilot monitoring will announce "1,000" and "500" **if no autocallout**

When FMS inbound course is captured

Approach FMS 1(2)

Deselect VNAV

Minimum 3nm to FAF: Set SPD MAN and reduce speed to VAPP

Flaps 4(5), Before Landing Checklist

Speed Checked, Flaps 4 (5) Before Landing Checklist

Accomplish Before Landing Checklist

Missed approach altitude can be set once established -3.0 down

Missed Approach Altitude Set ____

____ Set

First positive inward motion of final approach course

Course Alive

Flaps on F-speeds

Flaps 1

Speed Checked Flaps 1

Flaps on F-speeds

Flaps 2

Speed Checked Flaps2

Minimum of 10nm from FAF

Minimum 5nm to FAF
Flaps on F-speeds

Gear down Flaps 3

Gear Down Speed Checked Flaps 3

. 3 nm prior to FAF: FPA set to charted angle

FPA

FAF name Altitude

Checked

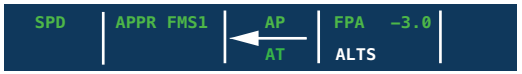
1,000ft AFE

Stable or Unstable Go Around

Stable

500 ft AFE

Stable or Unstable Go Around





DDA: Pilots must calculate and use Derived Decision Altitude (DDA) by adding 50 ft to the MDA. This is to ensure that the aircraft does not descend below the MDA during the go-around maneuver.

Go Around

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items

Go-around
Flap 2(4)

Press either TOGA button

Verify or move thrust levers to MAX

Flap 2(4)

Verify positive rate of climb

Positive Rate

Gear Up

Gear Up

Adjust speed to **VGA**

Verify or call the appropriate lateral modes

400 ft

HDG or FMS 1 (2)

Autopilot on

If desired

At or above 400' AGL

Adjust speed MAN SPD to 180 kts

(V)FLC

Acceleration Altitude

Crew May Either

Maintain current flap 2 configuration or accelerate and clean up

After Go Around Checklist

After Go Around Checklist complete





Landing FLAP	Go-Around FLAP
4	2
5	4

Single Engine Go Around

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items

Go-around
Flap 2(4)

Select TOGA

Flap 2(4)

Verify positive rate of climb

Positive Rate

Gear Up

Gear Up

Adjust speed to VAC

Verify or call the appropriate lateral modes

400 ft

HDG or FMS 1 (2)

Autopilot on

If desired

At or above 400' AGL

Set speed above green dot

1000'
(Or acceleration altitude)

Acceleration Altitude

Retract Flaps on Schedule

Flap 3/2/1

Speed Checked, Flap #

Flap 0, After Go Around Checklist

Speed Checked, Flap 0 After Go Around Checklist complete

After flaps 0, set green dot speed for climb





Landing FLAP	Go-Around FLAP
4	2
5	4

Discontinued Approach

Given an ATC Assigned Heading and Altitude

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent Action Items

ATC Clearance Example:

"MXY 203, cancel approach clearance, turn right heading 270, climb and maintain 8,000"

Discontinue Approach

Press ALT Button

Set assigned heading
Press HDG Button

Set assigned altitude

Select the most appropriate vertical mode (FPA, VS, or FLCH)

Verify, heading, altitude, appropriate lateral and vertical modes are selected and or engaged

Retract Flaps on Schedule

Flap 3/2/1

Speed Checked, Flap #

Flap 0, After Go Around Checklist

Speed Checked, Flap 0 After Go Around Checklist complete

Verify positive rate of climb

Positive Rate

Gear Up

Gear Up

Adjust speed **As Appropriate**



Above 2,100' AFE

The typical 400' and 1000' foot cues may not be applicable as the discontinued approach may commence above 1000' AFE.

Note:

If vectored for immediate return: Flaps and speed at pilots discretion

If vectored for a hold or diversion: Clean up aircraft to flaps 0

Landing FLAP	Go-Around FLAP
4	2
5	4

During the initial phase of the approach, there may be situations where the approach needs to be discontinued above **2,100 AFE (due to latched PFCC control).** A go around with TOGA can lead to excess thrust that may result in overshooting the missed approach altitude. Industry data suggests the go around maneuver is also a leading cause of undesirable aircraft state and flap overspeed events. For this situation, a discontinued approach is recommended.

Balked Landing

Initiate Go-Around After Main Gear Touchdown

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items

Note: After reverse thrust has been initiated, a full stop landing **MUST** be completed

Go-around

Press either TOGA button

Override thrust levers by pushing both to MAX

PF MUST press one or both disconnect buttons on the throttle quadrant after thrust is verified MAX

Monitor thrust and verify MAX thrust is achieved

When Thrust is at MAX

Thrust Set

Rotate to follow Flight Director OR towards 10 degrees if not present

Speed VGA with aircraft climbing

Flaps 2(4)

Flaps 2(4)

Positive rate of climb

Positive Rate

Gear Up

Gear Up

Verify SPD MAN and adjust to VGA

At or above 400' AGL

400 ft

Verify or call the appropriate lateral modes

HDG or FMS 1 (2)

Autopilot on

If desired

Acceleration Altitude

Adjust speed as required or select FMS speed.

(V)FLC

Retract flaps on schedule

Flap 3/2/1

Speed Checked Flap #

Flap 0, After Go Around Checklist

Speed Checked, Flap 0 After Go Around Checklist complete

Autothrottle may be re-engaged

CAUTION: DO NOT re-engage the autothrottle before selecting (V)FLC

Execute Go-Around procedure as per received clearance or ATC instructions

NOTE: CONFIG FLAP and CONFIG STAB may trigger. The Flight Crew must disregard these alerts

Landing FLAP	Go-Around FLAP
4	2
5	4

Clean Configuration Stall at Low Altitude

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent Action Items

Stall Configuration Set Up
Gear Up/ Flaps 0
Autopilot ON/ Autothrottle OFF
Thrust IDLE

Autopilot ON or OFF at instructors/check pilot discretion

Recovery Technique:
1. **Pitch:** Reduce AOA
2. **Roll:** Roll wings level
3. **Power:** Thrust as required

Called by the pilot first noticing the stall situation

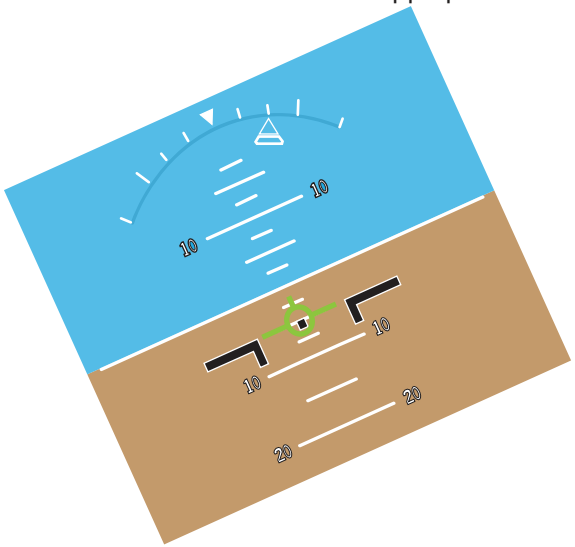
Stall

PM Duties
Checks AT and AP disengagement
Monitors altitude and speed

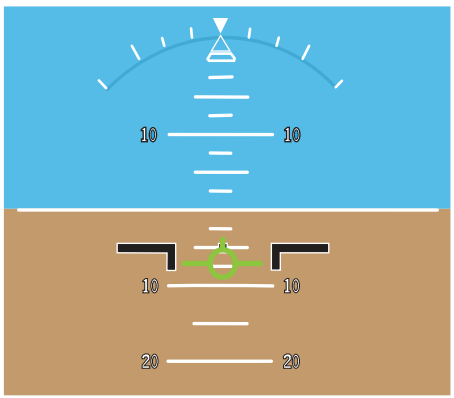


Pitch
Reduce AOA
Autopilot OFF
Autothrottle OFF

Pitch airplane reference symbol below the horizon as appropriate



Roll
Wings level

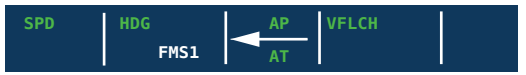


Power
Apply THRUST as required
Accelerate airplane to safe speed
Retract spoilers (sweep)

Control pitch in a smooth steady manner, respecting stick shaker

Speed Trim
As appropriate

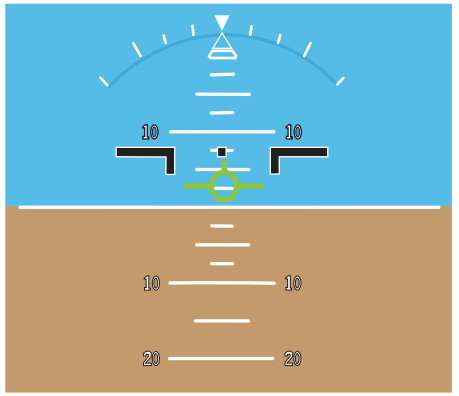
Once established at the recovery altitude and airspeed:



Repopulate the FMA

Recommended Modes
HDG or NAV
VFLC
AT ON
AP ON

Establish a positive rate of climb back to assigned altitude and course or HDG



Clean Configuration Stall at High Altitude

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent Action Items

Stall Configuration Set Up
Gear Up/ Flaps 0
Autopilot ON/ Autothrottle OFF
Thrust IDLE

Autopilot ON or OFF at instructors/check pilot discretion

Recovery Technique:
1. **Pitch:** Reduce AOA
2. **Roll:** Roll wings level
3. **Power:** Thrust as required

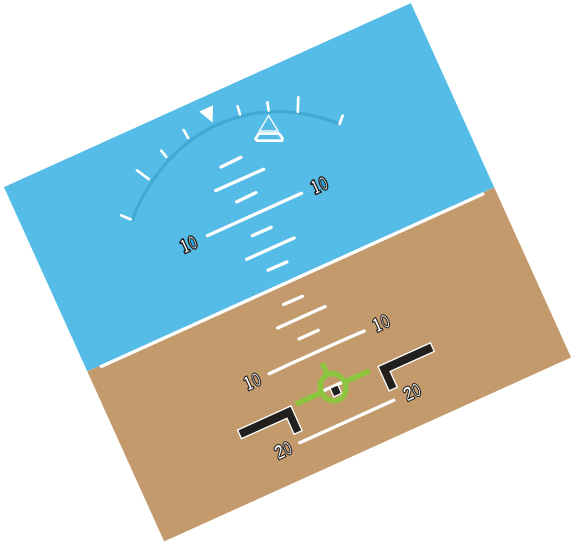
Called by the pilot first noticing the stall situation

Stall

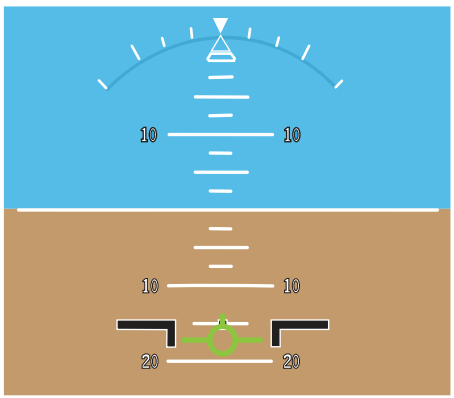
PM Duties
Checks AT and AP disengagement
Monitors altitude and speed



Pitch
Reduce AOA
Autopilot OFF
Autothrottle OFF
Pitch airplane reference symbol below the horizon as appropriate



Roll
Wings level

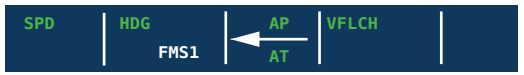


Power
Apply THRUST as required
Accelerate airplane to safe speed
Retract spoilers (sweep)

Control pitch in a smooth steady manner, respecting stick shaker

Speed Trim
As appropriate

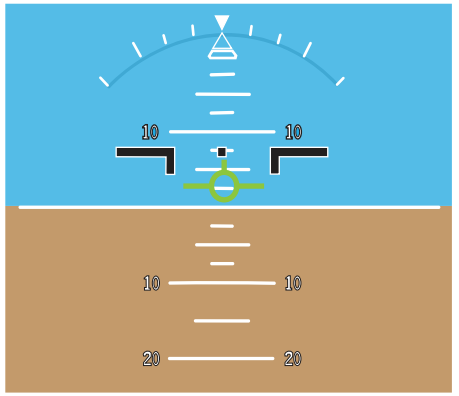
Once established at the recovery altitude and airspeed:



Repopulate the FMA

Recommended Modes
HDG or NAV
VFLC
AT ON
AP ON

Establish a positive rate of climb back to assigned altitude and course or HDG



Departure Configuration Stall

Breeze

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent Action Items

Stall Configuration Set Up
Gear Up/ Flaps 2
Autopilot ON/ Autothrottle OFF
Thrust IDLE
Turning @ 20 degrees bank

Autopilot ON or OFF at instructors/check pilot discretion

Recovery Technique:
1. **Pitch:** Reduce AOA
2. **Roll:** Roll wings level
3. **Power:** Thrust as required

Recommended Target Recovery Speeds
not a mandatory, graded speed

Transition to positive rate of climb: **150 kts**

Flaps 1: **170 kts**

Flaps 0: **190 kts**

Note: Continued acceleration required during cleanup

Called by the pilot first noticing the stall situation

Stall

Pitch
Reduce AOA
Autopilot OFF
Autothrottle OFF
Pitch airplane reference symbol below the horizon as appropriate

Power
Apply THRUST as required
Accelerate airplane to safe speed
Retract spoilers (sweep)

Roll
Wings level

Retract Flaps on Schedule

Flap 2/1

Speed Checked, Flap #

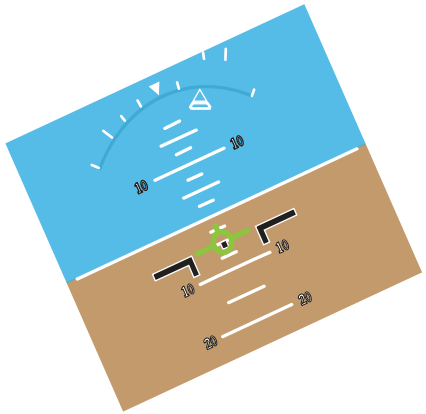
Speed Trim As appropriate

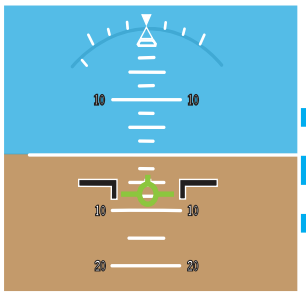
Flap 0, After Takeoff Checklist

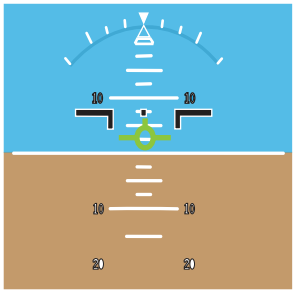
Speed Checked, Flap 0 After Takeoff Checklist Complete

Establish a positive rate of climb back to assigned altitude and course or HDG

Control pitch in a smooth steady manner, respecting stick shaker







Landing Configuration Stall

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent Action Items

Stall Configuration Set Up
Gear DN/ Flaps 4
Autopilot ON/ Autothrottle OFF
Thrust IDLE

Autopilot ON or OFF at instructors/check pilot discretion

Recovery Technique:
1. **Pitch:** Reduce AOA
2. **Roll:** Roll wings level
3. **Power:** Thrust as required

Recommended Target Recovery Speeds
not a mandatory, graded speed

Transition to positive rate of climb: **150 kts**

Flaps 1: **170 kts**

Flaps 0: **190 kts**

Note: Continued acceleration required during cleanup

Called by the pilot first noticing the stall situation

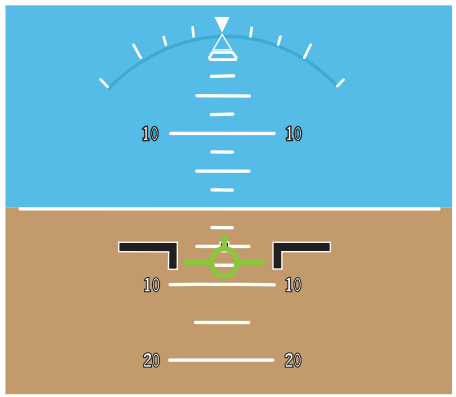
Stall

PM Duties
Checks AT and AP disengagement
Monitors altitude and speed



Pitch
Reduce AOA
Autopilot OFF
Autothrottle OFF

Pitch airplane reference symbol below the horizon as appropriate



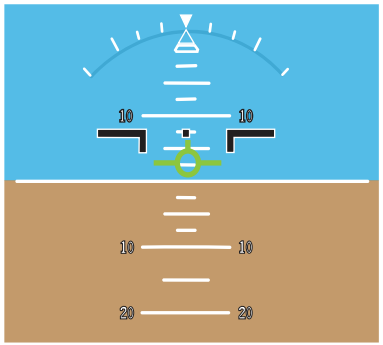
Roll
Wings level

Power
Apply THRUST as required
Accelerate airplane to safe speed
Retract spoilers (sweep)

Clean up aircraft on go around schedule

Establish a positive rate of climb back to assigned altitude and course or HDG

Control pitch in a smooth steady manner, respecting stick shaker



Windshear Prior to V1

Breeze

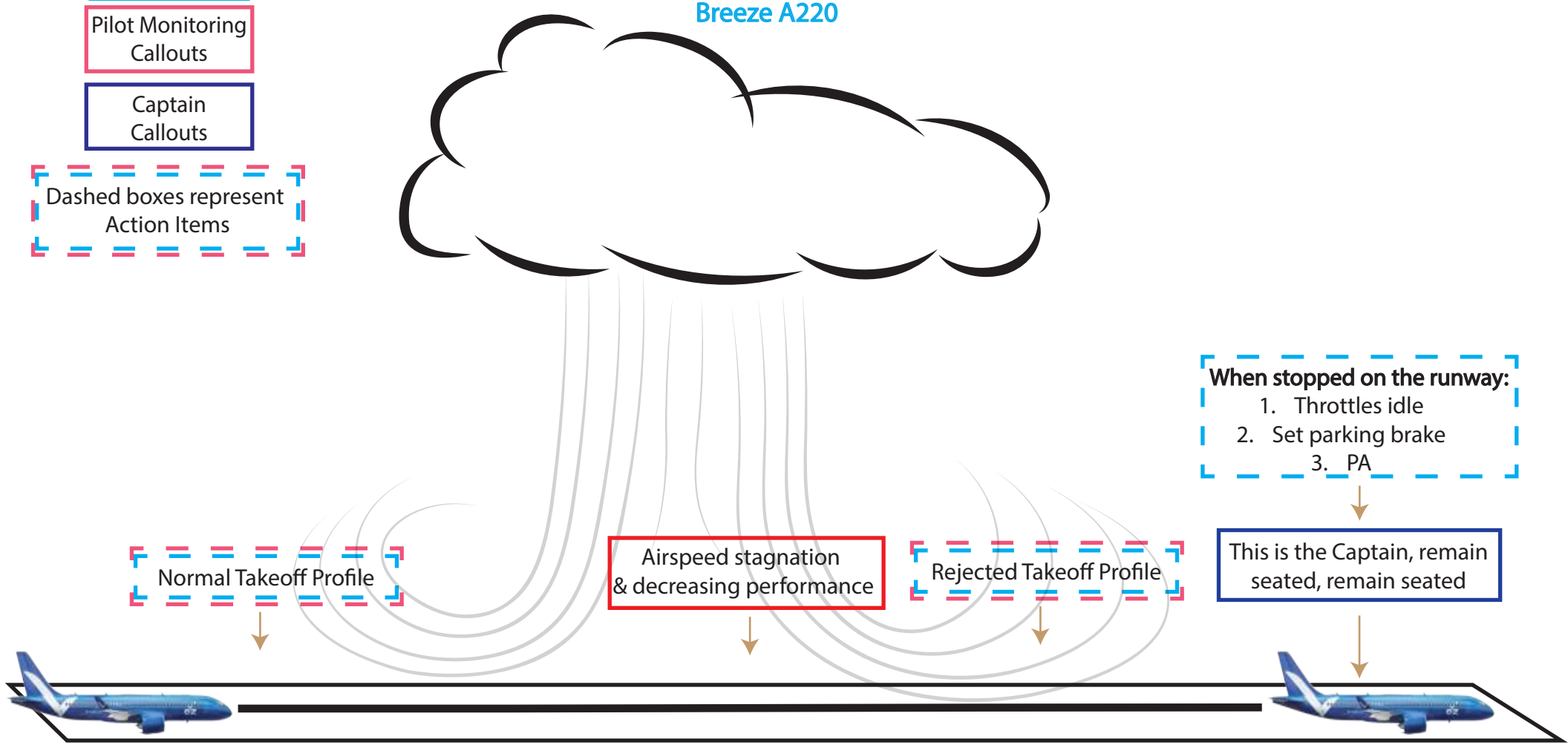
Pilot Flying Callouts

Pilot Monitoring Callouts

Captain Callouts

Dashed boxes represent Action Items

Breeze A220



Warning

A windshear encountered on the runway does not activate the windshear detection system. For detection, it is crucial to closely monitor airspeed during takeoff

Windshear After V1

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items



When clear of windshear

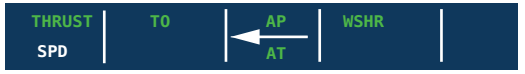
- Autopilot.....As required
- Autothrottle.....As required
- Landing Gear.....UP
- FD Modes.....As required
- FLAP.....Retract on schedule
- ATC.....Advise

When windshear is encountered, an aural, WINDSHEAR, WINDSHEAR, WINDSHEAR, WINDSHEAR indication sounds and a red WINDSHEAR message is shown on the PFD

Call out any significant variations in speed, rate of climb, pitch, thrust and attitude

When windshear warning is encountered

- Thrust Levers.....Advance to MAX
- TOGA.....Select
- FD Guidance.....Follow smoothly and precisely
- SPOILER lever.....Retract
- Configuration.....Maintain



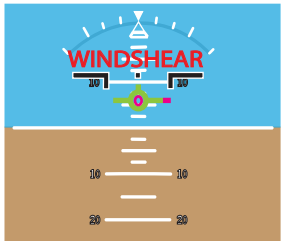
TO/WSHR (during takeoff) will stay active mode until another mode selection is made.

Takeoff Profile



Monitor flight instruments
Ensure proper thrust setting

Do not change landing gear or flap configuration until the vertical flight path is under control



NOTE: The airplane is equipped with a reactive Windshear Detection and Escape Guidance System that receives information from the TAWS. The system detects windshear conditions between 10 and 1500 feet radio altitude. The Detection and Escape Guidance System provides aural and visual alert messages and shows pitch and low speed cue symbology as guidance to escape/recover. If engaged, the **autopilot and autothrottle** can stay engaged.

Reactive Windshear During Approach

Breeze

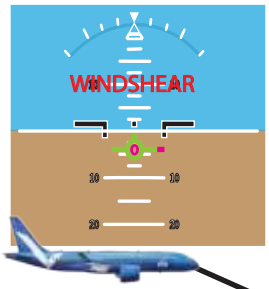
Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

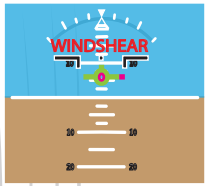
Dashed boxes represent Action Items

When windshear is encountered, an aural, WINDSHEAR, WINDSHEAR, WINDSHEAR, WINDSHEAR indication sounds and a red WINDSHEAR message is shown on the PFD



When windshear warning is encountered

Thrust Levers.....Advance to MAX
TOGA.....Select
FD Guidance.....Follow smoothly and precisely
SPOILER lever.....Retract
Configuration.....Maintain



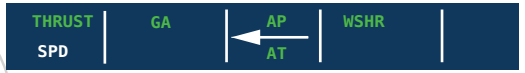
Monitor flight instruments
Ensure proper thrust setting

Do not change landing gear or flap configuration until the vertical flight path is under control

When clear of windshear

Autopilot.....As required
Autothrottle.....As required
Landing Gear.....UP
FD Modes.....As required
FLAP.....Retract on schedule
ATC.....Advise

Call out any significant variations in speed, rate of climb, pitch, thrust and attitude



GA/WSHR (during approach) will stay active mode until another mode selection is made.

In the absence of windshear FD escape guidance:

Thrust levers.....Advance to MAX
Sidestick.....Pull to the aft soft stop (respect the stick shaker)
Spoiler.....Retract
Configuration.....Maintain

NOTE: The airplane is equipped with a reactive Windshear Detection and Escape Guidance System that receives information from the TAWS. The system detects windshear conditions between 10 and 1500 feet radio altitude. The Detection and Escape Guidance System provides aural and visual alert messages and shows pitch and low speed cue symbology as guidance to escape/recover. If engaged, the **autopilot and autothrottle** can stay engaged.

NOTE: Gusty wind conditions not associated with downdrafts may occasionally trigger a windshear caution during an approach. If flight path is acceptable and the condition can be positively assessed as not related to thunderstorm or micro-burst activity, a go-around may not be required.

Cruise- OEI - Driftdown

Breeze A220

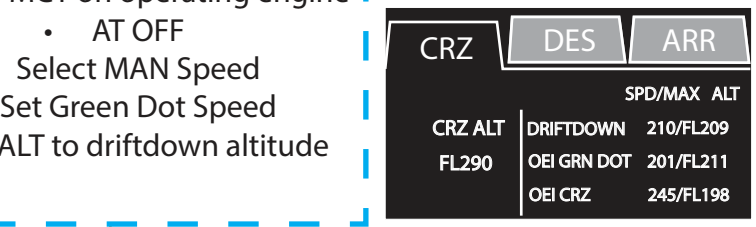
Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items

Unable to maintain altitude following loss of thrust

- Fly the airplane
- Verify MCT on operating engine
- AT OFF
- Select MAN Speed
- Set Green Dot Speed
- Set ALT to driftdown altitude



FMS OEI Section

If down line terrain is a factor, maintain the OEI green dot speed and altitude. If down line terrain is NOT a factor, descend to the desired altitude and establish the OEI cruise speed.

Remember - Since OEI cruise speed is faster than green dot speed, the altitude capability at the OEI cruise speed will be lower than the engine out ceiling obtained at the green dot speed.



Engine Failure

Engine Failure

Pilot first noticing the engine abnormal condition will state the condition

- Advise ATC
- Activation of **secondary flight plan** for escape route
- Monitor thrust, altitude and speed

Upon reaching green dot speed

FLC

Flight path stabilized

Reads and completes the applicable abnormal ECL

Remember - OEI green dot speed is NOT available until level at the driftdown

Level off at driftdown altitude

1. Following an engine failure or in-flight engine shutdown, an airplane may not be able to maintain its cruise altitude. The driftdown profile assures the airplane altitude is kept as high as possible throughout the descent. The driftdown speed is represented by the Green Dot and provides the lower descent gradient.
2. During a driftdown, the thrust increases as the airplane descends. Eventually, at a certain altitude the available thrust becomes equal to the airplane drag, and the airplane levels off. The altitude is called the gross level off altitude. The gross level off altitude, when corrected for gradient margins by 1.1% (two engines), this is called the net level off altitude.
3. Regulations require that the airplane be able to clear all terrain by a given margin when an engine fails. These calculations are a function of the dispatch planning software and will be noted in the Escape Routes section of your release. There are two methods of compliance for en route obstacle clearance that are allowed and used called method 1 or method 2.

Visual Approach

Breeze A220

Pilot Flying Callouts

Pilot Monitoring Callouts

Dashed boxes represent action Items

Visual Approach Rules: FCOM Chapter 3

1. Perform a visual approach at an altitude of 1,500' AFE or according to local authority regulations.
2. On VASI/PAPI equipped runways, maintain an altitude at or above the VASI/PAPI glidepath until a lower altitude is necessary for a safe landing.
3. If an instrument approach is available, pilots **must** back up the visual approach with an instrument approach.
4. FMS visual approaches can be loaded for any runway in the database. A waypoint 5nm (adjustable) from the runway is created with an altitude restriction of 1,500' AFE, creating a 3 degree vertical path to the runway.
5. During night visual approaches, alignment with the final approach course must be outside the FAF.

Entering Downwind

Speed 210

Flaps 1

Speed Checked Flaps 1

Abeam Threshold

Speed 180

Flaps 2

Speed Checked Flaps 2

Turning Base

Speed 160

Gear down Flaps 3

Gear Down Speed Checked Flaps 3



1,500' AFE

30 sec

Go-Around

Perform go-around profile

2nm

Base

Ensure SPD mode MAN
Adjust speed to a min VREF + 10

Flaps 4(5), Before Landing Checklist

Speed Checked, Flaps 4 (5) Before Landing Checklist

Accomplish Before Landing Checklist



Final

Set speed target VREF



700-500 AFE

Go-Around

A visual approach is not an instrument approach, therefore has no assigned missed approach segment. If a go-around is necessary climb to 1,500 AFE or as required to avoid terrain, obstacles, weather, or special use airspace.